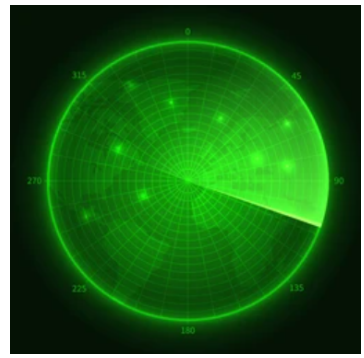
Problem A
Echolocation
Time limit: 5 seconds



One of the main challenges of deep sea navigation is the lack of light, rendering your crew vulnerable to unseen obstascles. However, your crew has risen to the challenge, having developed a hydrophone array to detect obstacles using echolocation. The system as of yet is still quite unreliable, though, so you'd like to devise an auxilliary system to detect errors in transmission.

The hydrophone array works by sending a series of pings, and collecting the signal from the resulting echoes. The echoes are characterized by k nonnegative integer values $A_0, A_1, \ldots, A_{k-1}$: when a ping is transmitted at time t, the hydrophone receives A_i units of signal at time $t + i$, for each $i = 0, 1, \ldots, k$. Any (integer) number of pings $x_t \geq 0$ may be transmitted at each nonnegative integer time t, and the total signal recorded at time j is the sum of contributions from all pings.

For example, suppose the echoes are characterized by $A_0 = 10$, $A_1 = 3$, $A_2 = 1$. Suppose you send $x_0 = 2$ pings at time $t = 0$, and $x_1 = 3$ pings at time $t = 1$. You will receive:

- Contribution from the 2 pings at time $t = 0$:

  $A_0 x_0 = 20$ units of signal at $t = 0$

  $A_1 x_0 = 6$ units of signal at $t = 1$

  $A_2 x_0 = 2$ units of signal at $t = 2$.

- Contribution from the 3 pings at time $t = 1$:

  $A_0 x_1 = 30$ units of signal at $t = 1$

  $A_1 x_1 = 9$ units of signal at $t = 2$

  $A_2 x_1 = 3$ units of signal at $t = 3$.

- Adding up the contributions from both sets of pings, the total signal that should be recorded is $W_0 = 20$, $W_1 = 36$, $W_2 = 11$, and $W_3 = 3$.

You have been given a recording of the total signal received at each observed time step (i.e. the observed W_j values). Determine whether there exists a valid (non-negative integer) sequence of ping transmission times that could produce this recording.

Input

The first line contains an integer k ($1 \leq k \leq 500$), the length of the echoes.

The second line contains k non-negative integers $A_0, A_1, \ldots, A_{k-1}$ ($0 \leq A_i \leq 10^9$). You are guaranteed that $A_0 \neq 0$ and $A_{k-1} \neq 0$.

The third line contains an integer T ($1 \leq T \leq 10^5$), the number of observed time steps.

The fourth line contains T non-negative integers $W_0, W_1, \ldots, W_{T-1}$ ($0 \leq W_i \leq 10^9$), the total signal received at each time step. You are guaranteed that $W_0 \neq 0$ and $W_{T-1} \neq 0$. No signal is received after time $T - 1$; that is, $W_i = 0$ for all $i \geq T$.

Output

Print YES if the observed recording is achievable, and NO otherwise.

Sample Input 1

```
3
10 3 1
4
20 36 11 3
```

Sample Output 1

```
YES
```

Sample Input 2

```
3
1 0 2
5
2 2 5 4 2
```

Sample Output 2

```
YES
```

Sample Input 3

```
3
2 536870912 2
3
536870912 0 536870912
```

Sample Output 3

```
NO
```

Sample Input 4

```
2
3 10
4
30 100 60 18
```

Sample Output 4

```
NO
```

Sample Input 5

```
5
2 5 8 7 6
4
2 5 8 7
```

Sample Output 5

```
NO
```

Problem B
Battleship
Time limit: 3 seconds

Admiral Clement Maximilian of the High Seas Patrol Command has found himself in a dire situation: His fleet is rapidly approaching a vast armada of enemy ships, spread out over a large section of ocean. Luckily, the Admiral has practiced for this very scenario, and he knows that if he strikes the ships in the right places, he can sink them all before they have a chance to retaliate! He has already successfully landed one strike on each ship, but he needs your help to finish the job. He has limited ammunition and can't afford to make many mistakes. Can you help the Admiral sink all the enemy ships without making too many incorrect guesses?

The Admiral uses an $n \times n$ grid to represent the ocean. Each cell of the grid is either empty or contains a part of a ship. All the enemy ships are the same length, l. The ships are located on the grid according to the following rules:

- Each ship occupies a contiguous sequence of l cells in either a horizontal or vertical direction.
- No two ships are adjacent to each other. This means that any two cells that share an edge cannot be occupied by different ships.

At each step, the Admiral will communicate to you the current state of the grid. The grid is represented by n lines of text, each containing a string of length n, composed of the characters `# . ?`. The c-th character of the r-th line of text represents the state of the cell in row r and column c:

- A `#` represents a cell that has been attacked (either initially by the Admiral, or later by you) and contains a ship.
- A `.` represents a cell that you have attacked that does not contain a ship. Whenever you attack a cell that does not contain a ship, this counts as a *miss*.
- A `?` represents a cell that has not been attacked yet, either by you or the Admiral.

Given one cell on each ship that the Admiral has already struck, make attacks to sink all the ships while making no more than $3k$ misses.

Interaction

This is an interactive problem.

The first line of input contains a single integer n ($1 \leq n \leq 40$), the size of the grid.

The second line of input contains a single integer k ($1 \leq k \leq n^2$), the number of ships.

The third line of input contains a single integer l ($2 \leq l \leq n$), the length of each ship.

The next n lines of input describe the initial state of the grid, as described above. All cells of this initial grid are either a `#` (attacked by the Admiral) or a `?` (not yet attacked). You are guaranteed that exactly one cell of each ship is a `#`.

To make an attack, output a single line of the format `? r c`, where r and c denote the row and column of the cell to attack. Since this is an interactive problem, you must flush the output stream after each line of output. This can be done with `fflush(stdout)` in C++, `System.out.flush()` in Java, or `sys.stdout.flush()` in Python.

After each attack, you will receive n lines of input, denoting the new state of the grid. That is, the character in row r and column c will be updated with the new status of that cell: either `#` if the attack hit a ship, or `.` if it missed.

You may make a maximum of $3k$ misses. If your program receives its $3k + 1$-st miss, it will be terminated and you will receive a verdict of `WRONG ANSWER`. You will also receive a `WRONG ANSWER` verdict if you make a guess out-of-bounds or in an incorrect format.

Once you have sunk all the ships, output a single line containing the character `!`. A ship is considered sunk when all the cells it occupies have been attacked (either by you, or by the Admiral's initial attacks). You will receive an `ACCEPTED` verdict if you have in fact sunk all the ships, and a `WRONG ANSWER` verdict otherwise.

Sample Interaction 1

Read	Write
5 2 3 ????? ????# ????? ?#??? ?????	
	? 1 5
????# ????# ????? ?#??? ?????	
	? 1 1
.???# ????# ????? ?#??? ?????	
	? 3 5
.???# ????# ????# ?#??? ?????	
	? 4 4
.???# ????# ????# ?#?#? ?????	
	? 4 3
.???# ????# ????# ?###? ?????	
	!

Problem C
Mariana Trench
Time limit: 1 second

Your team has mapped out the topography of the Mariana Trench, the deepest oceanic trench on Earth, stretching 2,550 kilometers across the western Pacific and extending down nearly 11 kilometers at its deepest point, Challenger Deep. The terrain along its length alternates between hadal sediment plains, basaltic ridges, hydrothermal fields, and exposed peridotite outcrops, but crucially, it follows regular, repeating patterns. Your team has decided to use lowercase letters to denote the different terrain types. Taking advantage of this regularity, your team has compressed their findings into a compact *string expression* rather than recording each terrain type individually.

Now, mission control has sent a query: given a position along the trench, what terrain type lies there? Decoding the full expression would take far too long. Can you efficiently recover just the character at a given index?

Formally, a *string expression* is defined recursively as follows:

- Any single lowercase letter is a string expression.
- If `X` and `Y` are string expressions, then `X+Y` is a string expression representing the concatenation of `X` and `Y`.
- If `X` is a string expression and `d` is a digit ($2 \le d \le 9$), then `(X)d` is a string expression denoting X repeated d times. Parentheses may be omitted when X is a single character.

Given a string expression and an index N, find the N-th character (1-indexed) of the string it represents.

Input

The first line contains the string expression. The string expression is guaranteed to have length between 1 and $4 \cdot 10^5$ inclusive.

The second line contains an integer N ($1 \le N \le 10^{17}$). It is guaranteed that the string represented by the given string expression has length at least N, but there are **no** guarantees on its maximum possible length.

Output

The N-th character of the represented string, 1-indexed.

Sample Input 1	Sample Output 1
`b+(((a+b)2+a)3)2` `1`	`b`

Sample Input 2	Sample Output 2
`b+(((a+b)2+a)3)2` `7`	`a`

Sample Input 3	Sample Output 3
`w5+r4+q9` `18`	`q`

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Problem D
Airlocks
Time limit: 2 seconds

You are designing a circular ring of deep-sea research laboratories, each operating at an integer pressure level. Your team has already decided how many labs they need at each pressure level to run their experiments; your job is just to assign pressure levels to each lab.

However, there's a catch: you forgot to order your custom high-quality airlocks! You'll have to settle for the cheap ones you have laying around. These airlocks assume a pressure differential of exactly 1. If the pressure differential is greater than that, or if it is zero, the airlock will malfunction and the whole setup will be ruined! This means you need to assign pressure levels such that any two adjacent labs' pressure levels differ by *exactly* 1.

How many distinct arrangements of pressure levels around the circle satisfy all these constraints? Since this number may be very large, output it modulo $10^9 + 7$. The labs are labeled, so rotations of a valid solution *are* considered distinct.

Input

The first line of input consists of a single integer n ($2 \leq n \leq 2 \cdot 10^5$).
The second line of input contains n space-separated integers $A_1, A_2, \ldots, A_n$ ($1 \leq A_i \leq 2 \cdot 10^5$), where A_i is the required number of labs at pressure level i.

Output

A single integer: the number of valid arrangements modulo $10^9 + 7$.

Sample Input 1	Sample Output 1
3 1 2 1	4

Sample Input 2	Sample Output 2
4 8 1 1 8	0

Sample Input 3	Sample Output 3
5 100 196 144 169 121	88807495

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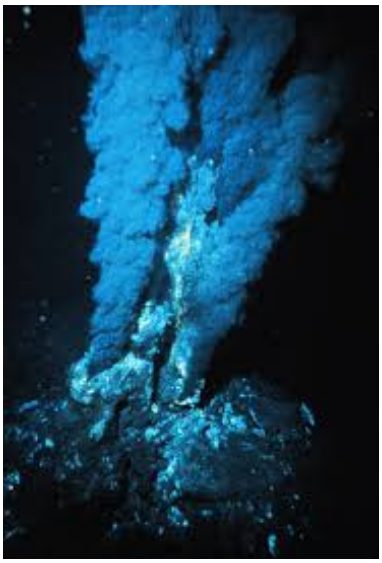
Problem E
Hydrothermal Vents (Div. 1)
Time limit: 5 seconds



You've been exploring the deep sea, and you have one final task before returning to your base: collecting a sample from one of the hydrothermal vents scattered throughout the terrain. Your crew has provided you with a handy map of the terrain, as well as instructions for your final mission.

The seafloor terrain is represented as a rectangular grid, with cells of the following types:

`A`	Start (your submarine)
`B`	Base (your destination)
`^`	Hydrothermal vent
`#`	Impassable rock
`.`	Open water

You may move one cell at a time in the four cardinal directions (up, down, left, right), but not diagonally. You are allowed to revisit cells: any cell other than `#` may be traversed any number of times.

Your route must satisfy all of the following:

- Starting at `A`, the path must end at `B`.
- The path must pass through at least one hydrothermal vent (`^`).
- The path may never pass through rock (`#`).
- The path may never go out of bounds of the grid.

Find the length of the shortest path that satisfies these conditions.

Input

The first line of input contains two space-separated integers n, m representing the number of rows and columns of the grid, respectively. You are guaranteed $1 \leq n, m \leq 2000$.

The next n lines of input each contain a string of length m, representing that row of the grid. You are guaranteed that the grid contains exactly one `A`, exactly one `B`, and at least one `^`. A valid path from `A` to `B` is guaranteed to exist.

Each cell is one of: `.` (open water), `#` (rock), `A` (start), `B` (base), or `^` (hydrothermal vent).

Output

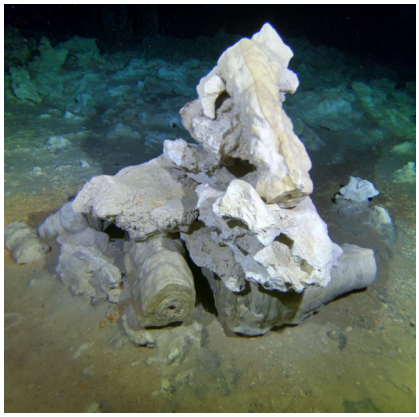
Output a single integer representing the minimum number of moves needed to get from the `A` to the `B` while reaching at least one hydrothermal vent. The path is allowed to cross over itself: any cell other than a `#` may be traversed an unlimited number of times.

Sample Input 1	Sample Output 1
4 7 ....A.^ .#####. B#####.	10

Sample Input 2	Sample Output 2
6 6 ^..... ###..^ ..^#.. A###.B ....^...	9

Problem F
Ariel Collection Management
Time limit: 3 seconds

Ariel Collection Management (ACM) is a firm specializing in maintaining and documenting underwater collections of human-origin objects (HOOs), including gadgets, gizmos, whosits, whatsits, and thingamabobs. ACM has recently been hired to catalog the collection of a high-ranking member of mermaid royalty, who wants to know which HOO she has the most of. You know that the only types of HOOs she has are the five listed above, and that no two types of object have the same count. Can you help ACM figure it out?



Input

The first line of input contains a single integer n ($10 \leq n \leq 2 \cdot 10^5$).

The next n lines of input each contain a single string, either `gadget`, `gizmo`, `whosit`, `whatsit`, or `thingamabob`, representing one of the objects in her collection. You are guaranteed that no two types of object occur the same number of times.

Output

Output two lines.

The first line of output should contain the name of the object type with the highest count (either `gadget`, `gizmo`, `whosit`, `whatsit`, or `thingamabob`).

The second line of output should contain its count, that is, how many times the most common object occurs in the collection.

Sample Input 1	Sample Output 1
10 gadget gizmo whosit whatsit gadget gizmo whosit gadget gizmo gadget	gadget 4

Sample Input 2	Sample Output 2
15 thingamabob whatsit whatsit whosit whosit whosit gadget gadget gadget gadget gizmo gizmo gizmo gizmo gizmo	gizmo 5

Problem G
Maximum Turmoil
Time limit: 2 seconds

Poseidon, ancient Greek god of the seas, has decided that there isn't enough turmoil in the water. He's had enough of these underwater dwellers and wants to make their journeys as chaotic as possible. To that end, he's decided to permanently change the currents of the oceans to cause maximum turmoil. However, having never taken a computer science course in his life, he needs your help figuring out what's the best way to do it.

The seas are divided up into n zones, labeled 1 through n. For each number i from 1 through n, you choose an integer a_i, denoting that water starting in zone i flows to zone a_i the next day. This process repeats every day: the ocean currents are relentless! However, each zone must receive water from exactly one other zone: otherwise, the seas would dry up.

For example, suppose there are 4 zones, and you assign $a_1 = 4, a_2 = 2, a_3 = 1, a_4 = 3$. Then, on the first day, the water in Zone 1 would flow to Zone 4, since $a_1 = 4$. On the second day, that water would flow to Zone 3, since $a_4 = 3$. On the third day, it would flow back to Zone 1, since $a_3 = 1$. The water in Zone 2 stays there indefinitely, since $a_2 = 2$.

It turns out that, no matter how the ocean currents are assigned, there will come a day where all the water is in their original zone at the same time. Define the *turmoil* of an assignment of ocean currents to be the number of days until this happens. For example, the turmoil is 3 for the example above. Given a value of n, can you find the maximum possible turmoil?

Input

Input will consist of a single integer, n, representing the number of zones.

Output

Output a single integer, the maximum possible turmoil of an assignment of ocean currents on n zones. This value is guaranteed to be less than 10^{18}.

Note: the answer for $n = 321$ exceeds 10^{18}, so you may assume that $1 \leq n \leq 320$.

Sample Input 1	Sample Output 1
4	4

Sample Input 2	Sample Output 2
9	20

Sample Input 3	Sample Output 3
320	941958815880242160

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Problem H
Gold and Silver
Time limit: 12 seconds

It's Captain Jack's lucky day! He's been searching all his life for abandoned shipwrecks full of great riches, and his dream has finally come true. In front of him are n treasure chests, each containing varying quantities of gold and silver coins. He needs to choose some chests to bring back onto his ship. Since he likes gold coins but hates anything silver, he wants the ratio $(\textit{gold coins})/(\textit{total coins})$ among all the coins he's bringing back to be as large as possible. However, if he does not bring back at least x coins in total, his crew will deem the mission a failure.

Find the maximum ratio $(\textit{gold coins})/(\textit{total coins})$ Captain Jack can achieve while collecting at least x coins in total.

Input

The first line of input contains an integer n ($1 \le n \le 40$), the number of treasure chests.

The second line of input contains an integer x ($1 \le x \le 10^9$) — the minimum number of coins required to bring back.

The third line of input contains n integers, where the i-th integer denotes the number of gold coins in the i-th container.

The fourth line of input contains n integers, where the i-th integer denotes the number of silver coins in the i-th container.

It is guaranteed that x does not exceed the total number of coins across all chests, and that the total number of coins across all chests does not exceed 10^9.

Output

Print a floating point number representing the maximum ratio. Your answer will be considered correct if its absolute or relative error does not exceed 10^{-9}.

Sample Input 1	Sample Output 1
3 2 1 4 1 0 3 1	0.66666666666666

Sample Input 2	Sample Output 2
4 22 3 1 4 1 2 8 1 2	0.4090909090909

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Problem I
Pronounceable Names
Time limit: 3 seconds

You and your commander have discovered a colossal, unexplored swath of the deep sea, and are logging all the new features and landmarks you're coming across. This means you get to name all these features. Unfortunately, your commander is insisting on completely unpronounceable names, such as `XNXAUUO`. No one's going to want to explore a land that they can't even pronounce! As the commander's subordinate, you don't want to change his names too much, but you at least want to make them pronounceable. So, you come up with some rules for what makes a pronounceable string.

A string of Latin letters is *pronounceable* if it never contains three consecutive vowels or three consecutive consonants. The vowels are `A, E, I, O, U`; all other letters are consonants. For simplicity, `Y` will always be considered a consonant.
Given an initial string of uppercase Latin letters, determine the minimum number of single-character replacements needed to make it pronounceable. A single-character replacement changes the letter at one position to a different letter, leaving all other positions unchanged.

Input

Input consists of a single string s of uppercase Latin letters of length n ($1 \leq n \leq 2 \cdot 10^5$).

Output

Output a single integer, the minimum number of characters that need to be changed of s to make it pronounceable.

Sample Input 1	Sample Output 1
`UVA`	`0`

Sample Input 2	Sample Output 2
`CHRISTMAAAAAAS`	`4`

Sample Input 3	Sample Output 3
`XNXAUUO`	`2`

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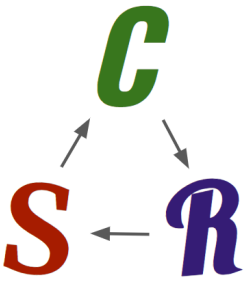
Problem J
Food Not-Chain (Div. 1)
Time limit: 12 seconds

This is an interactive problem.



In ecology, interactions between species don't necessarily have to fall into a strict food chain, or even a directed acyclic graph. In fact, we observe a clear cyclical relationship between three particular strains of *Escherichia Coli*: Strain C, a toxin-producing strain; Strain S, a toxin-sensitive strain; and strain R, a toxin-resistant strain.

- Strain C dominates Strain S, because Strain S is vulnerable to the Strain C toxin.
- Strain R dominates Strain C, because it is immune to the toxin.
- Strain S dominates Strain R, because it doesn't have to spend as many resources on toxin resistance.

You wonder to yourself: "Can I find one of these cycles in the wild"? Since there are so many possibilities, you consult your Ecology Specialist to choose just a few interesting species to consider. They decide on a set of n species, labeled 1 through n. They choose these n species such that the following conditions are satisfied:

- In every pair of different species, one dominates the other.
- Every species among these n dominates at least one other species.

Given this information, your goal is to determine whether there are three species among the ones that your Specialist selected with a cyclical dominance relationship. You can do this by asking your Specialist which of two species would dominate. However, you don't want to ask your Specialist too many questions: it takes a lot of time and effort to determine these dominance relationships! Can you find a 3-species cycle while asking no more than $3n$ questions, or prove that none exists?

Interaction

This problem uses multi-test.

The first line of input contains a single integer t $(1 \le t \le 2000)$, representing the number of test cases.

Each test case consists solely of a single integer n $(3 \le n \le 30000)$, representing the number of species.

To make a query, output a single line of the form `? a b`, where $1 \le a, b \le n$ and $a \ne b$. In response, you will receive a single line of input: either `a`, if species a dominates species b, or `b`, if species b dominates species a. Every species is guaranteed to dominate at least one other species among these n. Your program will terminate with a verdict of `WRONG ANSWER` if you exceed $3n$ queries on this test case.

To declare a 3-cycle, output a single line of the form `! a b c`, where $1 \le a, b, c \le n$, and a, b, c are all distinct. Your answer will be considered correct if Species a dominates Species b, Species b dominates Species c, and Species c dominates Species a, in which case interaction will proceed to the next test case. Otherwise, you will receive a verdict of `WRONG ANSWER`. Note that the order of a, b, c matters!

To declare that no 3-cycle exists among these n species, output a single line consisting of the following text: `! -1 -1 -1`. Your answer will be considered correct if there do not exist a, b, c where pecies a dominates Species b, Species b dominates Species c, and Species c dominates Species a.

You should make exactly one declaration (either of a 3-cycle, or that none exists) per test case, after making no more than $3n$ queries.

It is guaranteed that the sum of n over all test cases does not exceed 30000.

Read	Sample Interaction 1	Write
2 3		
		? 2 1
1		
		! 1 2 3
4		
		? 1 2
2		
		? 1 3
3		
		? 1 4
1		
		? 2 3
3		
		? 2 4
4		
		? 3 4
4		
		! 2 1 4

Problem K
The Kraken
Time limit: 5 seconds

The Kraken: a legendary, gigantic sea monster of Norwegian folklore, capable of creating whirlpools powerful enough to sink entire vessels. After reading the ancient texts, you've determined that the way to fend off the Kraken is with a zigzag shape, its sharp corners disturbing any semblance of its whirlpools' circular nature. Only a very particular zigzag setup will work, though. Can you create the exact shape you need and save your crew?

Your task is to print a zigzag pattern of width w and height h on a grid of dots (.) and hashes (#). The zigzag pattern you need satisfies the following conditions:

- The top-left cell is a #.
- Each column contains exactly one #.
- A # in the top row has # neighbors diagonally below-left and below-right (provided that these locations stay within grid bounds).
- A # in the bottom row has # neighbors diagonally above-left and above-right (provided that these locations stay within grid bounds).
- All other # cells have exactly one diagonal neighbor above and exactly one diagonal neighbor below.

Input

A single line containing two integers w ($1 \leq w \leq 79$) and h ($2 \leq h \leq 79$).

Output

h lines of exactly w characters each, forming the zigzag pattern.

Sample Input 1	Sample Output 1
12 3	#...#...#... .#.#.#.#.#.# ..#...#...#.

Sample Input 2	Sample Output 2
17 5	#.....#.....# .#.....#.#.....#. ..#...#...#...#.. ...#.#.....#.#... ....#.....#....

Sample Input 3	Sample Output 3
4 7	#... .#.. ..#. ...# ....

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Problem L
The Leviathan
Time limit: 2 seconds

The Leviathan is a mythical dragon-like sea monster symbolizing primordial chaos: impenetrable shields for scales, breath that sets coals ablaze, smoke pouring from its nostrils. Ancient texts describe it as incomprehensibly fearsome, where mere mortals have no hope of even coming close to getting it under control. You and your crew know that you have no chance of slaying the beast, and your only hope is to find a way to avoid it altogether. Luckily, after thousands of years of studying it, you've cracked just a little bit about its movement pattern. Can you use this to determine whether you're safe from its wrath?

You and the Leviathan both occupy cells on an m-by-n grid. Your position is fixed, but the Leviathan may move if it desires. It moves two spaces diagonally: if it starts at cell (r, c), it can move to the cell $(r \pm 2, c \pm 2)$, skipping over the intervening cells, provided that the destination cell is still within bounds of the grid. Your task is to determine whether the Leviathan can make a sequence of zero or more moves in succession to end up at exactly your position.

Input

Input consists of a single line. That line will contain 6 integers: m, n, x_1, y_1, x_2, y_2. The value m denotes the number of rows in the grid, and n denotes the number of columns in the grid. Your position on the grid is given by (x_1, y_1), where rows and columns are 1-indexed; the Leviathan's starting position is given by (x_2, y_2). It is guaranteed that $1 \le x_1, x_2 \le m \le 10^9$, and $1 \le y_1, y_2 \le n \le 10^9$.

Output

Output `YES` if the Leviathan can reach your position in zero or more moves. Otherwise, output `NO`.

Sample Input 1	Sample Output 1
`5 7 1 1 3 7`	`YES`

Sample Input 2	Sample Output 2
`3 3 1 1 1 2`	`NO`

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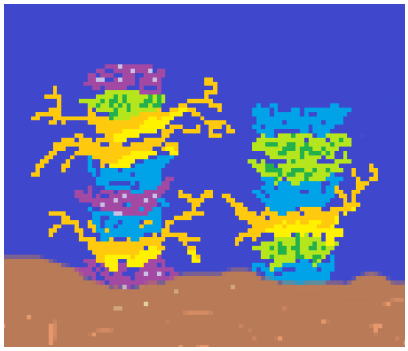
Problem M
Stacked Coral
Time limit: 3 seconds



Your research vessel has been exploring the coral reefs on the seafloor. You've located two natural columns of stacked coral specimens, each composed of four species: Aquamarine (`A`), Byzantium (`B`), Cerulean (`C`), and Dandelion (`D`). Your team wants to consolidate them into a single large column, and has already determined a specific target arrangement of species that they want to end up with. They've also prepared a separate (initially empty) site for the final stack, where the coral specimens can be safely placed. However, to avoid damaging the fragile coral specimens, only the following move is permitted:

- At each step, you can move the top coral specimen from one of the two columns to the top of the final stack.

You can successively perform moves of this type until there are no more coral specimens in either of the initial stacks. Under these constraints, can you achieve the target arrangement? If so, how?

Input

Input consists of three lines. On each of these lines, the first character of the string represents the bottom of the stack, and the last character represents the top.

The first line contains a non-empty string of characters representing the Column 1 starting arrangement. This string has length a $(1 \leq a \leq 4000)$, and consists only of the uppercase characters `A B C D`.

The second line contains a non-empty string of characters representing the Column 2 starting arrangement. This string has length b $(1 \leq b \leq 4000)$, and consists only of the uppercase characters `A B C D`.

The third line contains a non-empty string of characters representing the target arrangement. This string is guaranteed to have length $a + b$, and consists only of the uppercase characters `A B C D`.

Output

If it is possible to build the target arrangement under these constraints, output a string of `1`s and `2`s of length $a + b$, where the i-th character indicates which column to take from on the i-th step. If it is impossible, output the string `IMPOSSIBLE`.

If multiple valid sequences exist, output any one of them.

Sample Input 1	Sample Output 1
ADBB BCBCC CCBBBDACB	222111122

Sample Input 2	Sample Output 2
ABA AAB BBAAAA	IMPOSSIBLE

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